



OAK RIDGE NATIONAL LABORATORY

Tracking the Power of Water: The U.S. Hydropower Market Report

Hydropower is a vital component of the US renewable energy landscape, contributing nearly 30% of the nation's renewable electricity in 2022. As the push for decarbonization intensifies, hydropower's role in providing reliable, flexible power to support intermittent sources like wind and solar has become increasingly crucial. Despite hydropower's importance, however, comprehensive data on the US fleet's performance, development trends, and challenges were historically lacking, leaving policymakers and industry stakeholders without reliable information to guide the decision-making process.

To address this need, the US Department of Energy (DOE) in 2015 commissioned the U.S. Hydropower Market Report, filling a critical void in publicly available data. Compiled and published by DOE's Oak Ridge National Laboratory (ORNL), the HMR is the only regularly updated source of comprehensive information on the status and direction of US hydropower. The report tracks activities across the life cycle of hydropower plants—from new project permitting and construction to license transfers and performance trends—and serves as an essential reference for the hydropower community, including industry stakeholders and policymakers.

A Decade of Data: Revealing Trends and Key Impacts

The HMR is grounded in data from a combination of public and commercial databases. Since the HMR's inception, four full reports have been published—in 2015, 2018, 2021, and 2023—with updated summary slides available annually. Data used to produce plots in the reports and slides are also published and accessible through ORNL's HydroSource. Following are some of the key findings from the 2023 report:

- **Capacity Growth:** From 2010 to 2022, the US conventional hydropower fleet grew by 2.1 GW, reaching a total capacity of 80.58 GW across 2,252 plants. Growth was driven mainly by upgrading existing plants and adding hydropower at non-powered dams and conduits.
- **Pumped Storage Hydropower (PSH) Trends:** The PSH fleet remains a cornerstone of the nation's energy storage capacity, providing 70% of utility-scale power storage capacity and 96% of energy storage. The US added 1.4 GW of PSH capacity from 2010 to 2022, with 97% of this increase coming from upgrades to existing facilities.



KEY TRENDS IN HYDROPOWER



CONTINUED CAPACITY
GROWTH



NEW CAPACITY FROM
DAM RETROFITS



IMPROVED FLEET
RELIABILITY

- **Development Pipeline:** By the end of 2022, 117 new hydropower facilities with a combined capacity of 1.2 GW were being actively considered for development. 95% of this proposed new capacity would come from non-powered dam retrofits. The PSH development pipeline included an additional 96 projects totaling 91 GW, though none had begun construction.
- **Performance and Reliability:** Hydropower generation has declined slightly because of drought conditions in the West, with a median capacity factor of 35.3% in 2020–2022, down from 38.8% in the 2010s. After more than a decade of slow but steady decreases, the availability factor, which depends on the number of hours a hydropower unit spends in repairs and forced outages, remained stable in 2019–2021, reflecting the fleet’s continued reliability.

Charting the Future of Hydropower

The HMR serves as an unbiased reference on US hydropower development, prices, and trends for industry stakeholders, federal partners, and policymakers. The report informs strategic planning and policy proposals, providing essential data to illustrate the value, challenges, and opportunities of hydropower in the United States. The report also guides DOE hydropower R&D planning and investment by the Water Power Technologies Office.

Beyond its role in policy and strategy, the HMR is a vital public outreach tool. It improves general understanding of hydropower and its critical role in decarbonizing the US electric grid. This outreach is essential for fostering informed discussions, public awareness, and support for hydropower initiatives. By providing clear and accessible information, the HMR helps guide policymakers and industry partners in making decisions that ensure the continued growth and reliability of this vital energy resource.

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